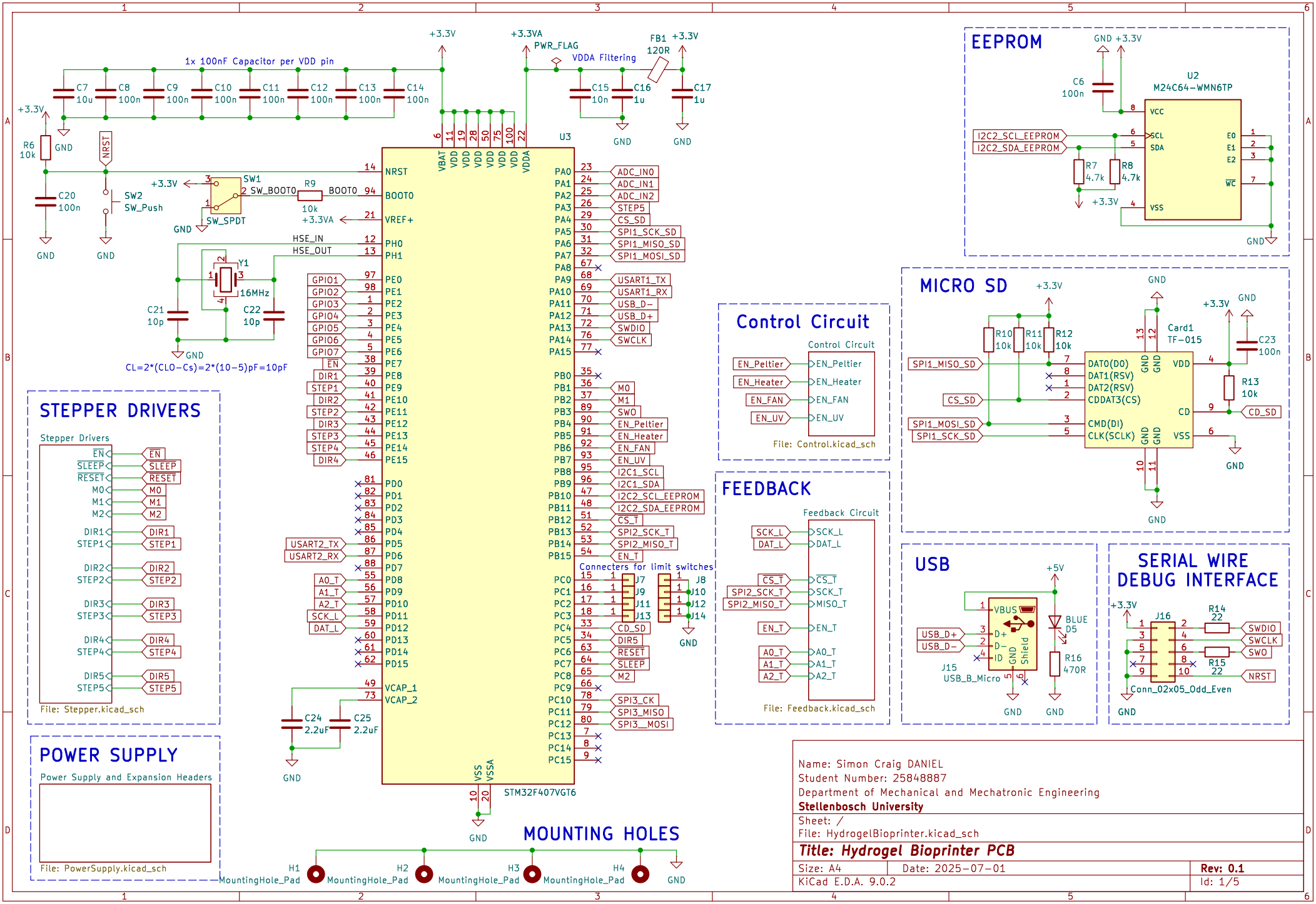
1x 100nF Capacitor per VDD pin
C7 10u
C8 100n
C9 100n
C10 100n
C11 100n
C12 100n
C13 100n
C14 100n
+3.3V
+3.3VA
PWR_FLAG
VDDA Filtering
FB1 120R
C15 10n
C16 1u
C17 1u
GND
R6 10k
NRST
C20 100n
SW2 SW_Push
SW1 SW_SPDT
R9 10k
BOOT0
VREF+
HSE_IN
HSE_OUT
Y1 16MHz
C21 10p
C22 10p
CL=2*(CLO-Cs)=2*(10-5)pF=10pF
U3
VBAT
VDD
VDDA
VSS
VSSA
NRST
BOOT0
VREF+
PH0
PH1
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VCAP_1
VCAP_2
C24 2.2uF
C25 2.2uF
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
STM32F407VGT6
ADC_IN0
ADC_IN1
ADC_IN2
STEP5
CS_SD
SPI1_SCK_SD
SPI1_MISO_SD
SPI1_MOSI_SD
USART1_TX
USART1_RX
USB_D-
USB_D+
SWDIO
SWCLK
M0
M1
SWO
EN_Peltier
EN_Heater
EN_FAN
EN_UV
I2C1_SCL
I2C1_SDA
I2C2_SCL_EEPROM
I2C2_SDA_EEPROM
CS_T
SPI2_SCK_T
SPI2_MISO_T
EN_T
CD_SD
DIR5
RESET
SLEEP
M2
SPI3_CK
SPI3_MISO
SPI3__MOSI
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
EN
DIR1
STEP1
DIR2
STEP2
DIR3
STEP3
DIR4
STEP4
USART2_TX
USART2_RX
A0_T
A1_T
A2_T
SCK_L
DAT_L
Connecters for limit switches
J7
J8
J9
J10
J11
J12
J13
J14
MOUNTING HOLES
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
EEPROM
U2
M24C64-WMN6TP
C6 100n
R7 4.7k
R8 4.7k
VCC
SCL
SDA
VSS
E0
E1
E2
WC
Control Circuit
Control Circuit
File: Control.kicad_sch
FEEDBACK
Feedback Circuit
File: Feedback.kicad_sch
MICRO SD
Card1 TF-015
R10 10k
R11 10k
R12 10k
R13 10k
C23 100n
DAT0(DO)
DAT1(RSV)
DAT2(RSV)
CDDAT3(CS)
CMD(DI)
CLK(SCLK)
VDD
CD
VSS
USB
J15 USB_B_Micro
VBUS
D+
D-
ID
Shield
BLUE D5
R16 470R
SERIAL WIRE DEBUG INTERFACE
J16
Conn_02x05_Odd_Even
R14 22
R15 22
STEPPER DRIVERS
Stepper Drivers
File: Stepper.kicad_sch
POWER SUPPLY
Power Supply and Expansion Headers
File: PowerSupply.kicad_sch
Name: Simon Craig DANIEL
Student Number: 25848887
Department of Mechanical and Mechatronic Engineering
Stellenbosch University
Sheet: /
File: HydrogelBioprinter.kicad_sch
Title: Hydrogel Bioprinter PCB
Size: A4
Date: 2025-07-01
Rev: 0.1
KiCad E.D.A. 9.0.2
Id: 1/5